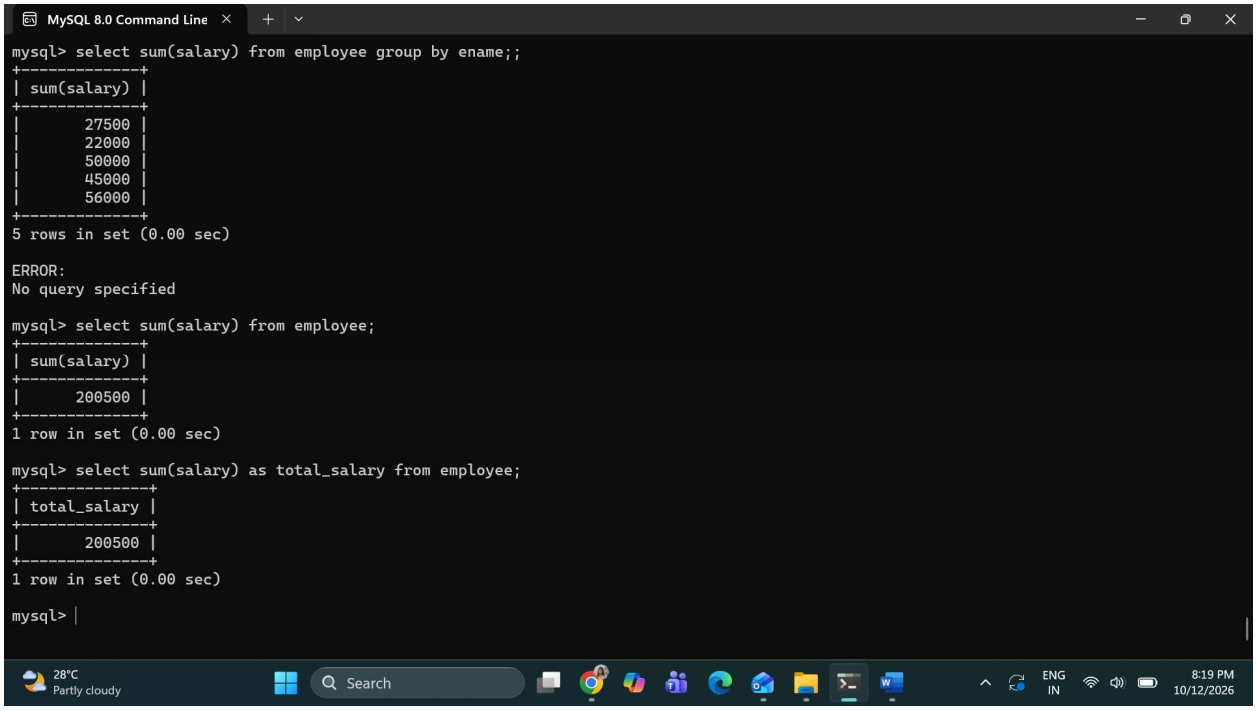
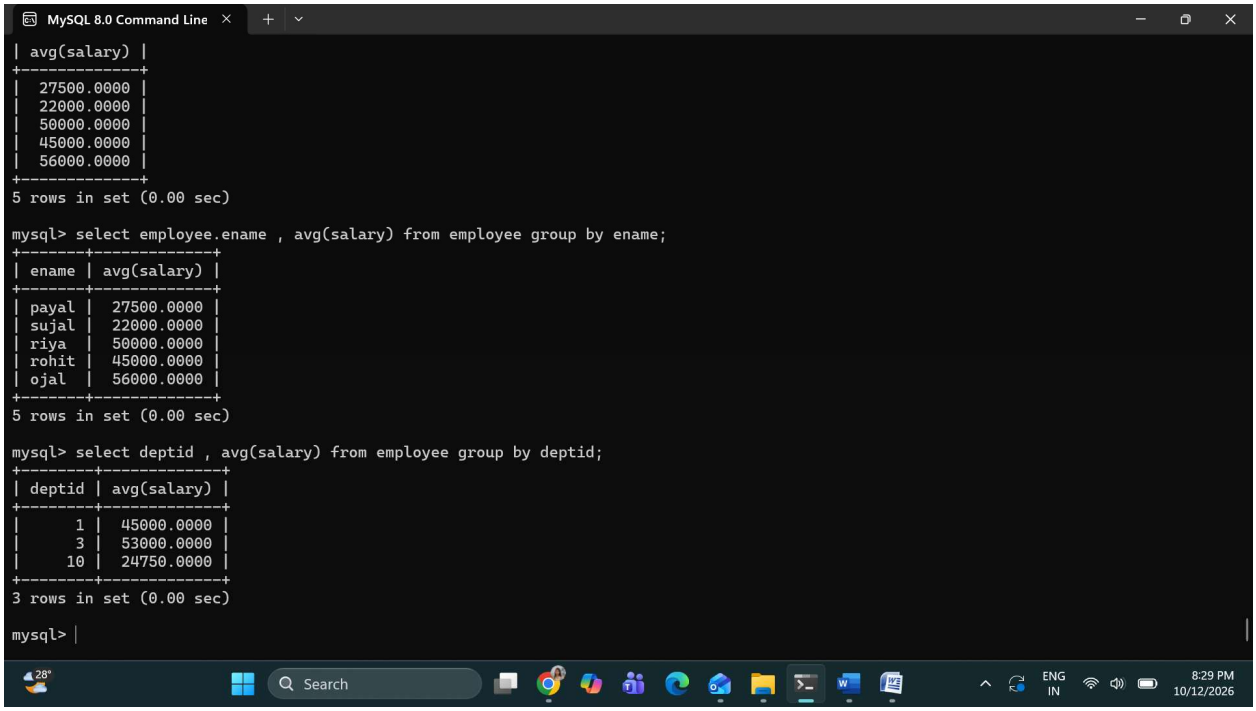
Assignment 3

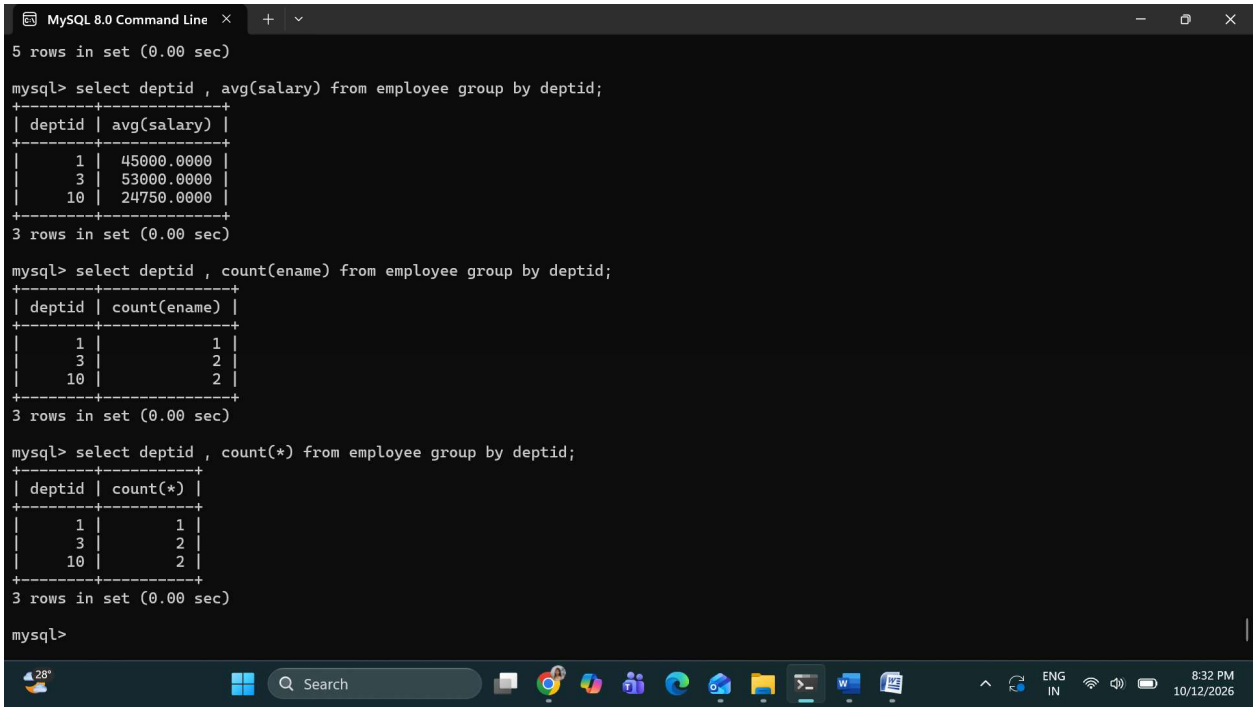
Q1.



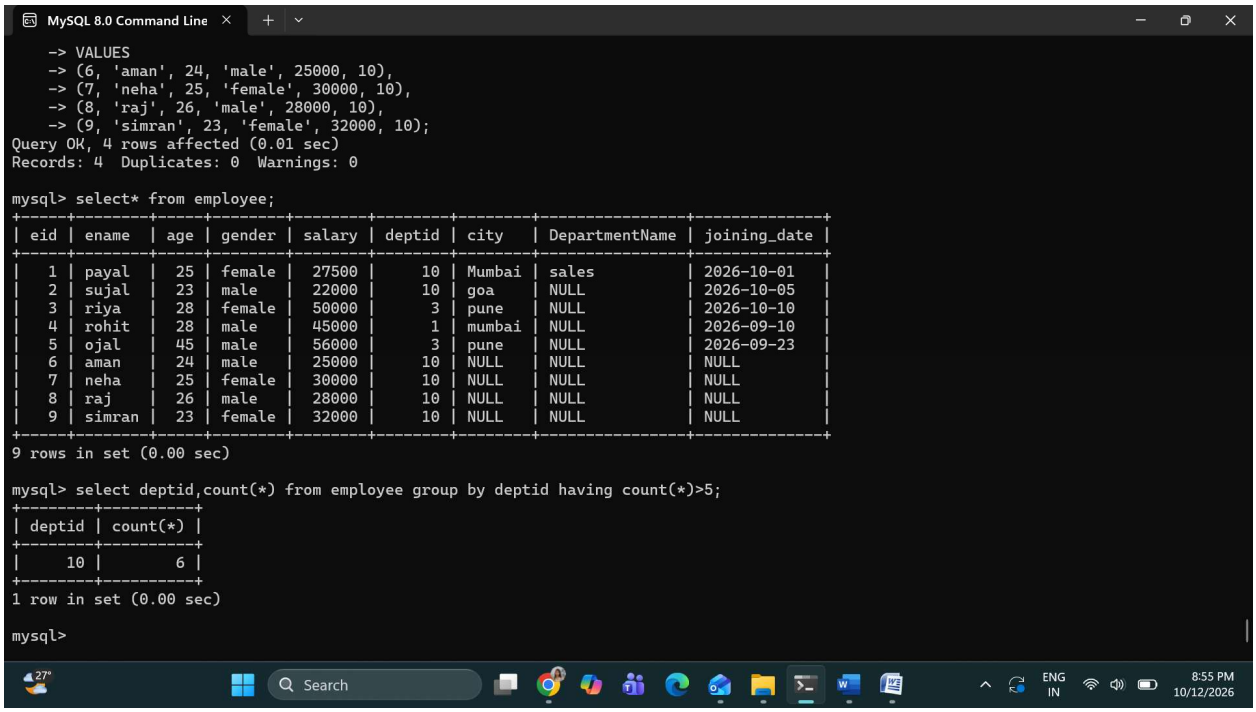
Q2.



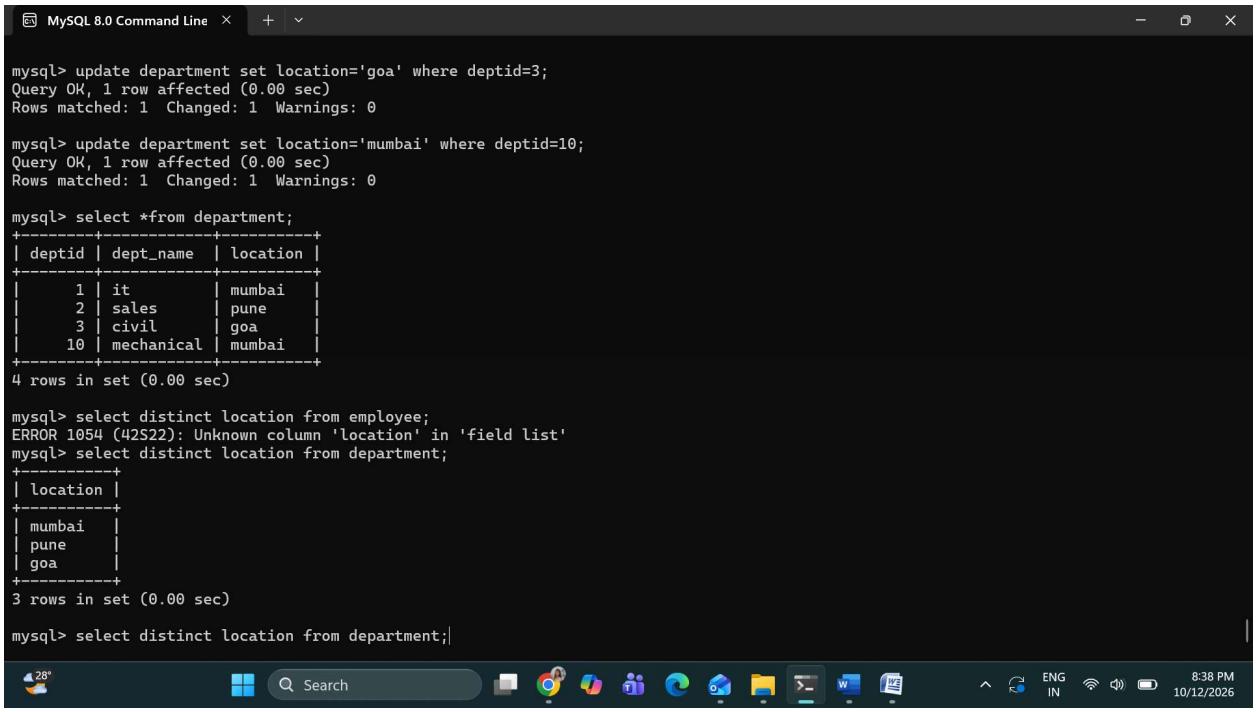
Q3.


MySQL 8.0 Command Line
5 rows in set (0.00 sec)
mysql> select deptid , avg(salary) from employee group by deptid;
deptid	avg(salary)
1	45000.0000
3	53000.0000
10	24750.0000
3 rows in set (0.00 sec)
mysql> select deptid , count(ename) from employee group by deptid;
deptid	count(ename)
1	1
3	2
10	2
3 rows in set (0.00 sec)
mysql> select deptid , count(*) from employee group by deptid;
deptid	count(*)
1	1
3	2
10	2
3 rows in set (0.00 sec)
mysql>
Search
ENG IN
8:32 PM 10/12/2026

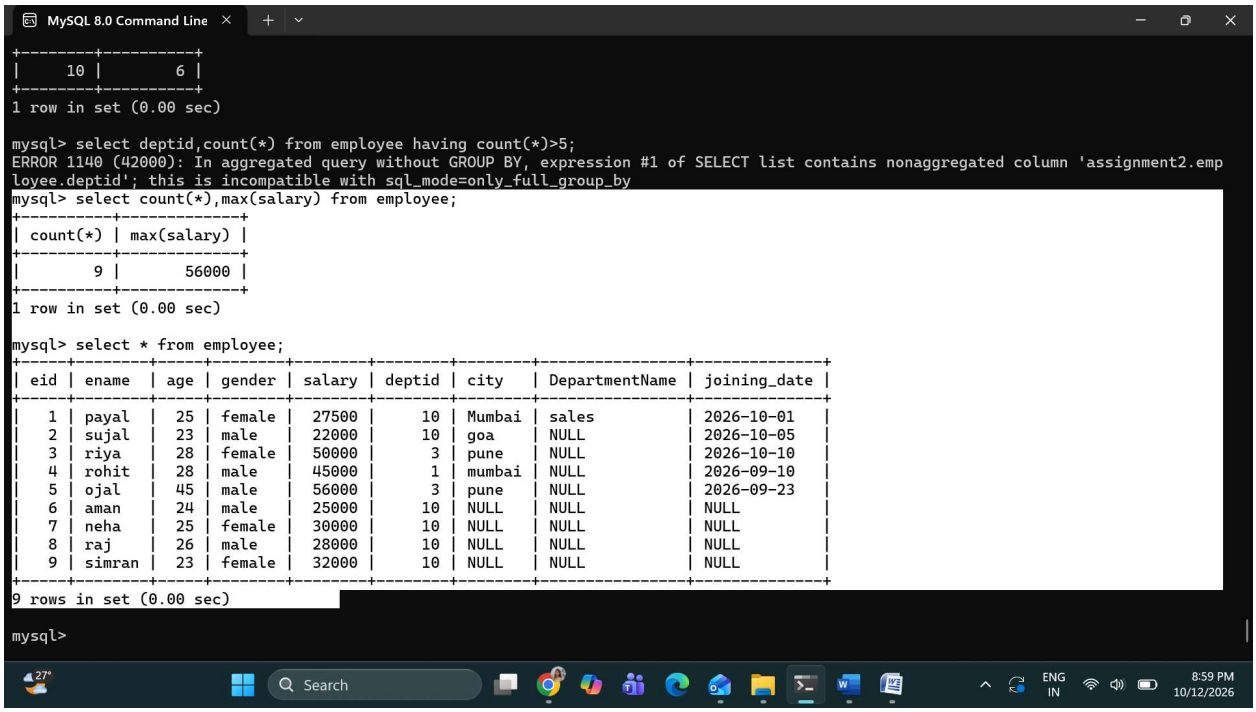
Q4.


MySQL 8.0 Command Line
-> VALUES
-> (6, 'aman', 24, 'male', 25000, 10),
-> (7, 'neha', 25, 'female', 30000, 10),
-> (8, 'raj', 26, 'male', 28000, 10),
-> (9, 'simran', 23, 'female', 32000, 10);
Query OK, 4 rows affected (0.01 sec)
Records: 4 Duplicates: 0 Warnings: 0
mysql> select* from employee;
eid	ename	age	gender	salary	deptid	city	DepartmentName	joining_date
1	payal	25	female	27500	10	Mumbai	sales	2026-10-01
2	sujal	23	male	22000	10	goa	NULL	2026-10-05
3	riya	28	female	50000	3	pune	NULL	2026-10-10
4	rohit	28	male	45000	1	mumbai	NULL	2026-09-10
5	ojal	45	male	56000	3	pune	NULL	2026-09-23
6	aman	24	male	25000	10	NULL	NULL	NULL
7	neha	25	female	30000	10	NULL	NULL	NULL
8	raj	26	male	28000	10	NULL	NULL	NULL
9	simran	23	female	32000	10	NULL	NULL	NULL
9 rows in set (0.00 sec)
mysql> select deptid,count(*) from employee group by deptid having count(*)>5;
deptid	count(*)
10	6
1 row in set (0.00 sec)
mysql>
Search
ENG IN
8:55 PM 10/12/2026

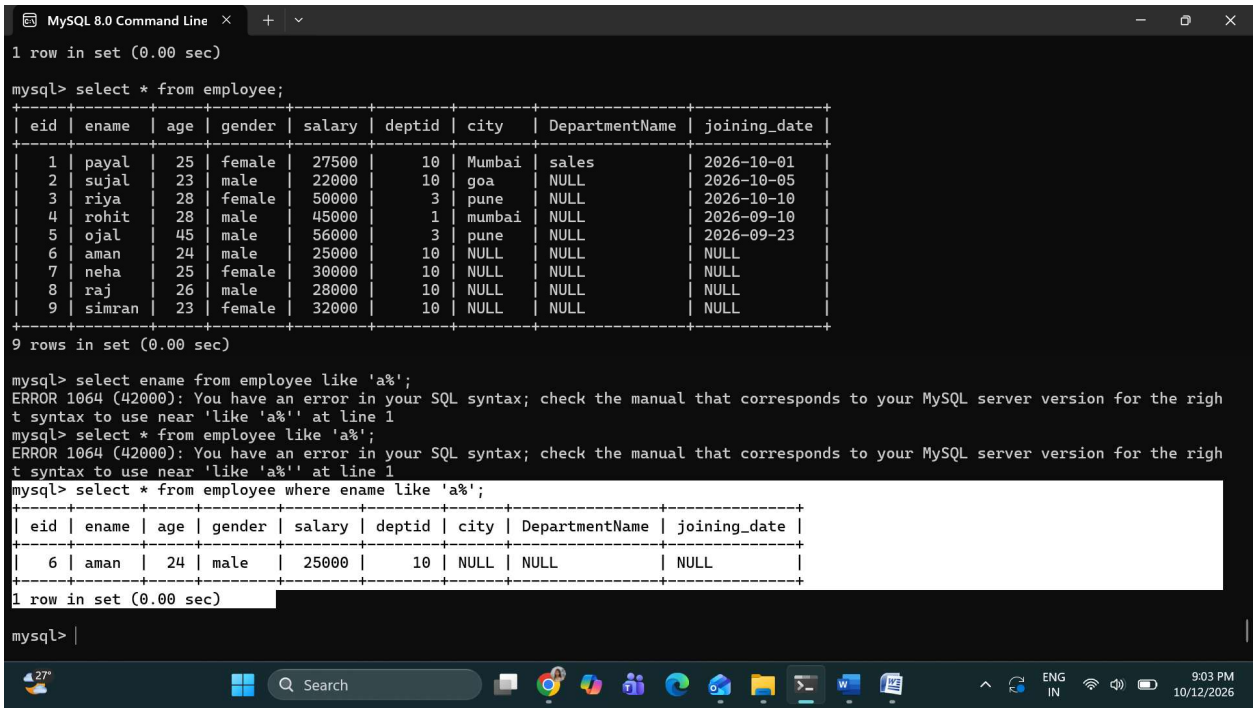
Q5.


MySQL 8.0 Command Line
mysql> update department set location='goa' where deptid=3;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0
mysql> update department set location='mumbai' where deptid=10;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0
mysql> select *from department;
+--------+------------+----------+
| deptid | dept_name | location |
+--------+------------+----------+
1	it	mumbai
2	sales	pune
3	civil	goa
10	mechanical	mumbai
+--------+------------+----------+
4 rows in set (0.00 sec)
mysql> select distinct location from employee;
ERROR 1054 (42S22): Unknown column 'location' in 'field list'
mysql> select distinct location from department;
+----------+
| location |
+----------+
| mumbai |
| pune |
| goa |
+----------+
3 rows in set (0.00 sec)
mysql> select distinct location from department;
Search
ENG
IN
8:38 PM
10/12/2026

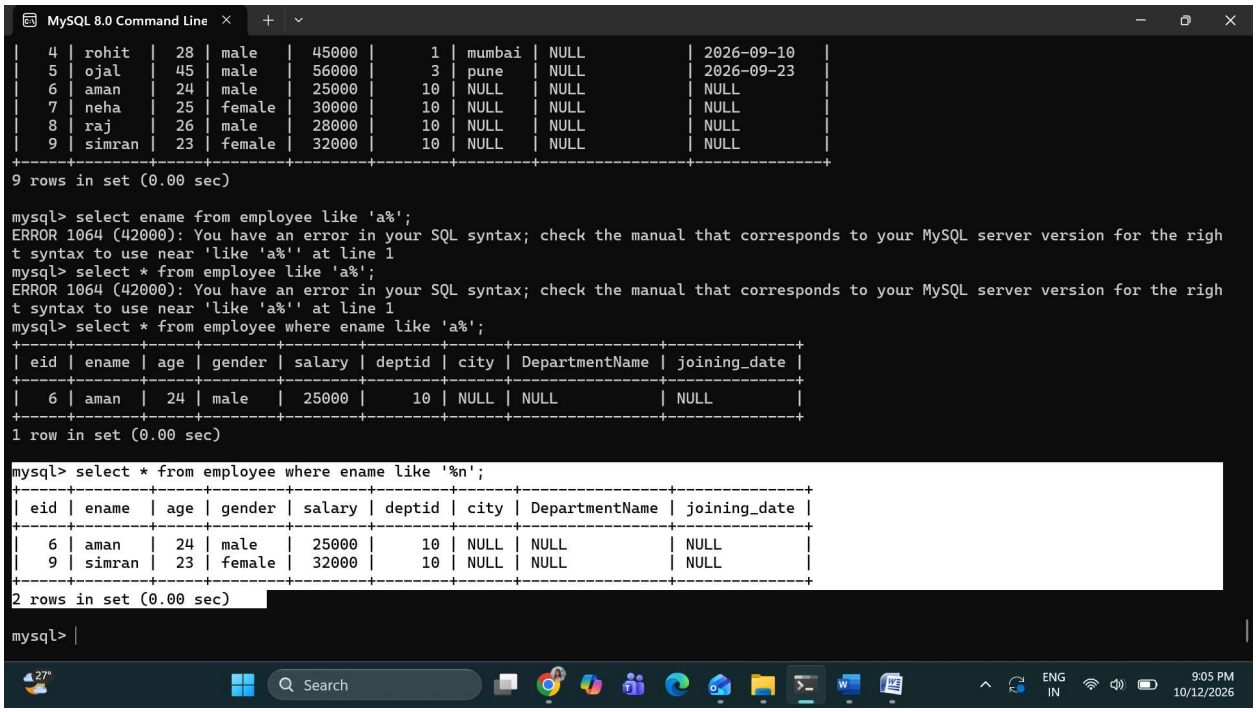
Q6.


MySQL 8.0 Command Line
+--------+----------+
| 10 | 6 |
+--------+----------+
1 row in set (0.00 sec)
mysql> select deptid,count(*) from employee having count(*)>5;
ERROR 1140 (42000): In aggregated query without GROUP BY, expression #1 of SELECT list contains nonaggregated column 'assignment2.emp
loyee.deptid'; this is incompatible with sql_mode=only_full_group_by
mysql> select count(*),max(salary) from employee;
+----------+-------------+
| count(*) | max(salary) |
+----------+-------------+
| 9 | 56000 |
+----------+-------------+
1 row in set (0.00 sec)
mysql> select * from employee;
+-----+--------+-----+--------+--------+--------+--------+----------------+--------------+
| eid | ename | age | gender | salary | deptid | city | DepartmentName | joining_date |
+-----+--------+-----+--------+--------+--------+--------+----------------+--------------+
1	payal	25	female	27500	10	Mumbai	sales	2026-10-01
2	sujal	23	male	22000	10	goa	NULL	2026-10-05
3	riya	28	female	50000	3	pune	NULL	2026-10-10
4	rohit	28	male	45000	1	mumbai	NULL	2026-09-10
5	ojal	45	male	56000	3	pune	NULL	2026-09-23
6	aman	24	male	25000	10	NULL	NULL	NULL
7	neha	25	female	30000	10	NULL	NULL	NULL
8	raj	26	male	28000	10	NULL	NULL	NULL
9	simran	23	female	32000	10	NULL	NULL	NULL
+-----+--------+-----+--------+--------+--------+--------+----------------+--------------+
9 rows in set (0.00 sec)
mysql>
Search
ENG
IN
8:59 PM
10/12/2026

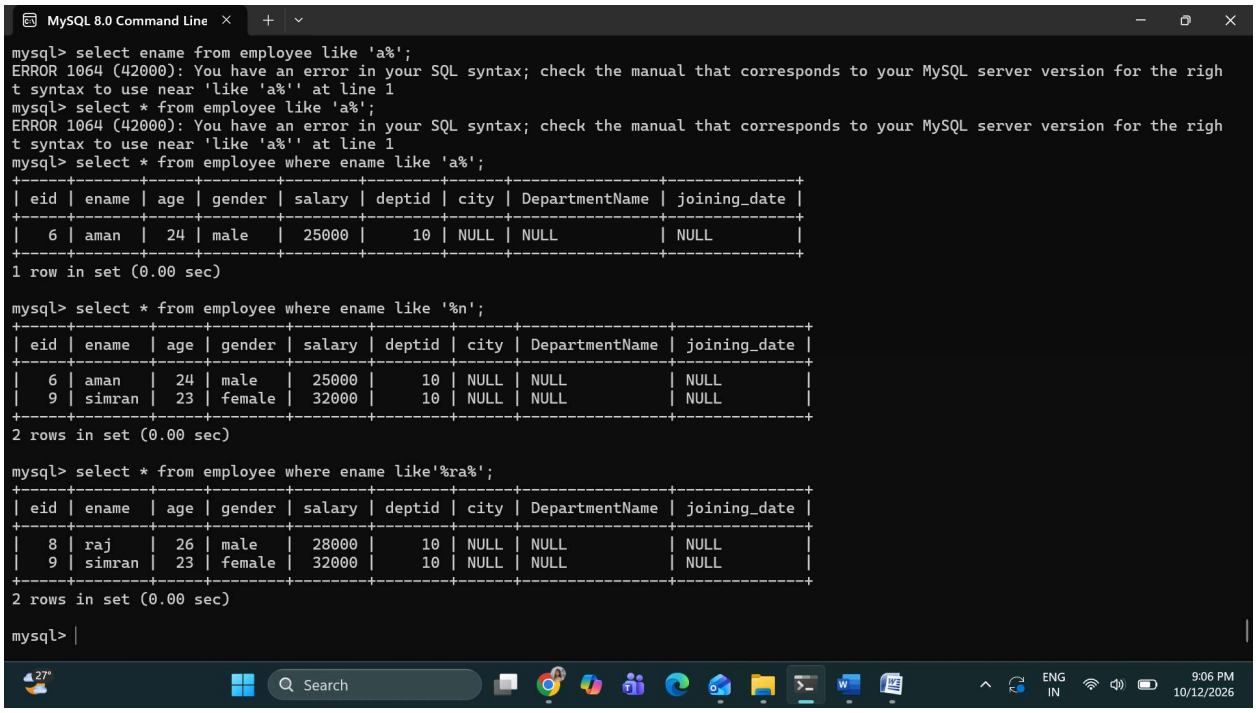
Q7.


MySQL 8.0 Command Line
1 row in set (0.00 sec)
mysql> select * from employee;
+-----+--------+-----+--------+--------+---------+--------+----------------+--------------+
| eid | ename | age | gender | salary | deptid | city | DepartmentName | joining_date |
+-----+--------+-----+--------+--------+---------+--------+----------------+--------------+
1	payal	25	female	27500	10	Mumbai	sales	2026-10-01
2	sujal	23	male	22000	10	goa	NULL	2026-10-05
3	riya	28	female	50000	3	pune	NULL	2026-10-10
4	rohit	28	male	45000	1	mumbai	NULL	2026-09-10
5	ojal	45	male	56000	3	pune	NULL	2026-09-23
6	aman	24	male	25000	10	NULL	NULL	NULL
7	neha	25	female	30000	10	NULL	NULL	NULL
8	raj	26	male	28000	10	NULL	NULL	NULL
9	simran	23	female	32000	10	NULL	NULL	NULL
+-----+--------+-----+--------+--------+---------+--------+----------------+--------------+
9 rows in set (0.00 sec)
mysql> select ename from employee like 'a%';
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the righ
t syntax to use near 'like 'a%'' at line 1
mysql> select * from employee like 'a%';
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the righ
t syntax to use near 'like 'a%'' at line 1
mysql> select * from employee where ename like 'a%';
+-----+-------+-----+--------+--------+---------+------+----------------+--------------+
| eid | ename | age | gender | salary | deptid | city | DepartmentName | joining_date |
+-----+-------+-----+--------+--------+---------+------+----------------+--------------+
| 6 | aman | 24 | male | 25000 | 10 | NULL | NULL | NULL |
+-----+-------+-----+--------+--------+---------+------+----------------+--------------+
1 row in set (0.00 sec)
mysql>
Search
ENG
IN
9:03 PM
10/12/2026

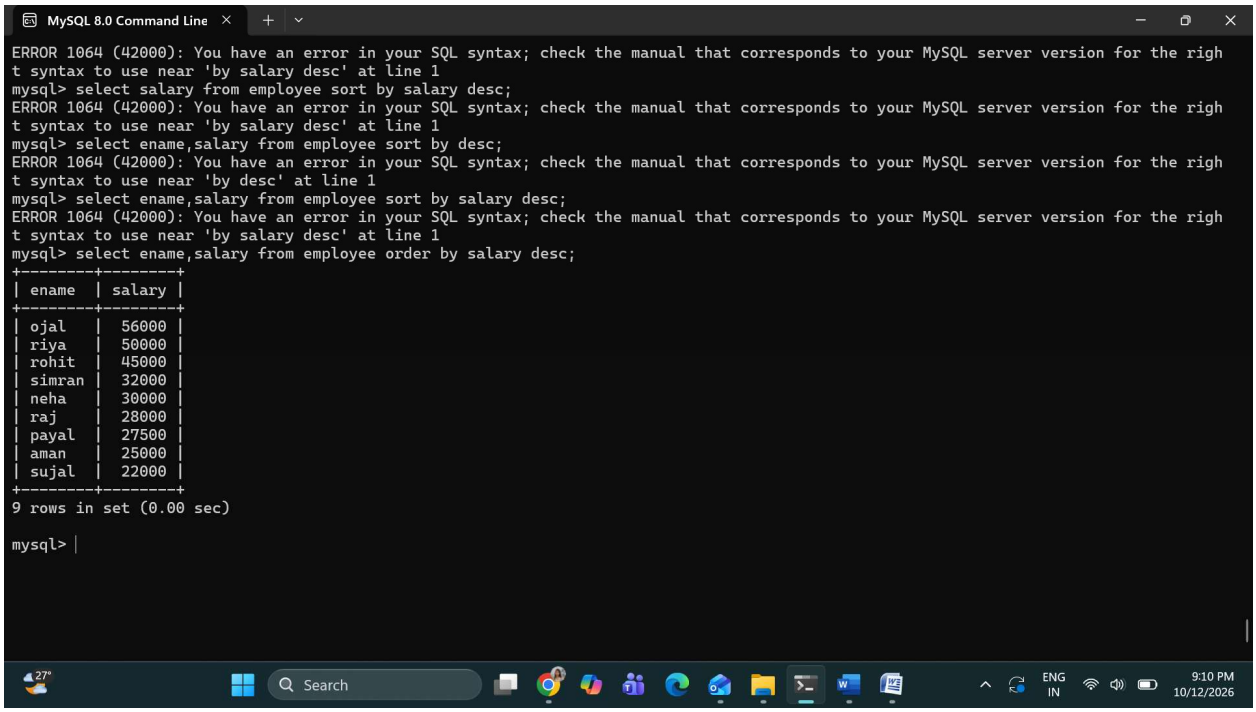
Q8.


MySQL 8.0 Command Line
4	rohit	28	male	45000	1	mumbai	NULL	2026-09-10
5	ojal	45	male	56000	3	pune	NULL	2026-09-23
6	aman	24	male	25000	10	NULL	NULL	NULL
7	neha	25	female	30000	10	NULL	NULL	NULL
8	raj	26	male	28000	10	NULL	NULL	NULL
9	simran	23	female	32000	10	NULL	NULL	NULL
9 rows in set (0.00 sec)
mysql> select ename from employee like 'a%';
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the righ
t syntax to use near 'like 'a%'' at line 1
mysql> select * from employee like 'a%';
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the righ
t syntax to use near 'like 'a%'' at line 1
mysql> select * from employee where ename like 'a%';
6	aman	24	male	25000	10	NULL	NULL	NULL
1 row in set (0.00 sec)
mysql> select * from employee where ename like '%n';
6	aman	24	male	25000	10	NULL	NULL	NULL
9	simran	23	female	32000	10	NULL	NULL	NULL
2 rows in set (0.00 sec)
mysql>
Search
ENG
IN
9:05 PM
10/12/2026

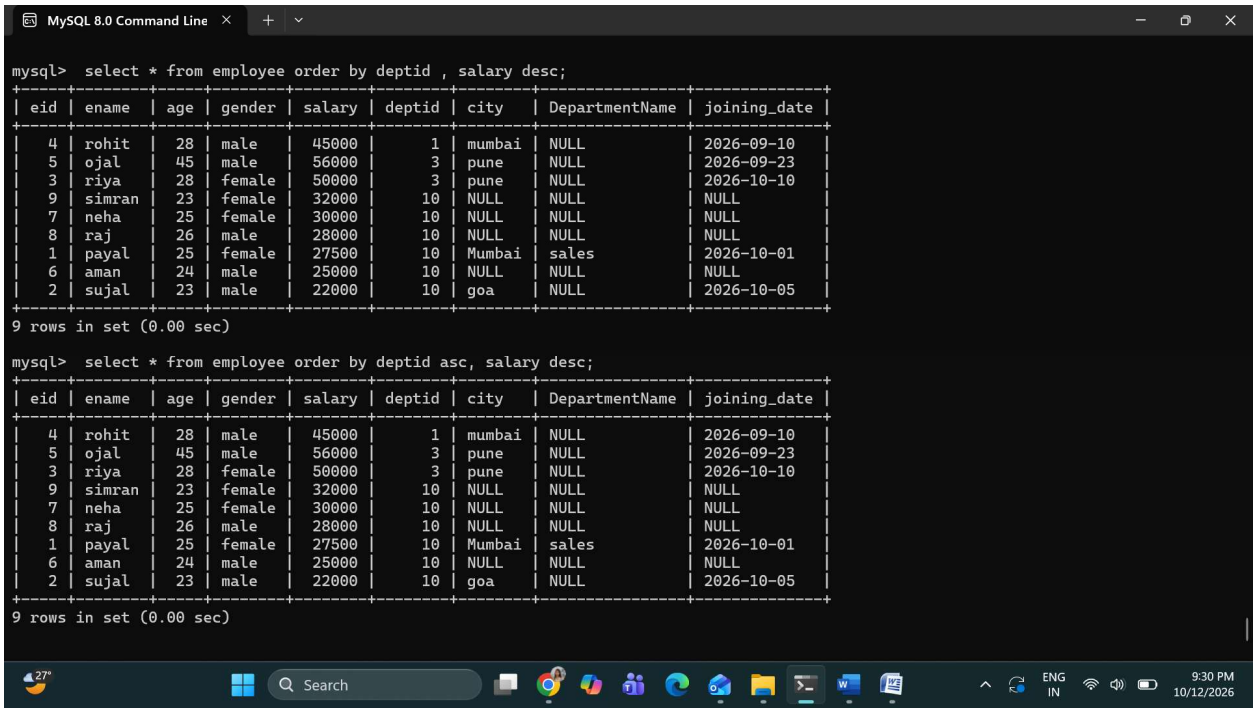
Q9.


MySQL 8.0 Command Line
mysql> select ename from employee like 'a%';
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the righ
t syntax to use near 'like 'a%'' at line 1
mysql> select * from employee like 'a%';
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the righ
t syntax to use near 'like 'a%'' at line 1
mysql> select * from employee where ename like 'a%';
| eid | ename | age | gender | salary | deptid | city | DepartmentName | joining_date |
| 6 | aman | 24 | male | 25000 | 10 | NULL | NULL | NULL |
1 row in set (0.00 sec)
mysql> select * from employee where ename like '%n';
eid	ename	age	gender	salary	deptid	city	DepartmentName	joining_date
6	aman	24	male	25000	10	NULL	NULL	NULL
9	simran	23	female	32000	10	NULL	NULL	NULL
2 rows in set (0.00 sec)
mysql> select * from employee where ename like'%ra%';
eid	ename	age	gender	salary	deptid	city	DepartmentName	joining_date
8	raj	26	male	28000	10	NULL	NULL	NULL
9	simran	23	female	32000	10	NULL	NULL	NULL
2 rows in set (0.00 sec)
mysql>
Search
ENG
IN
9:06 PM
10/12/2026

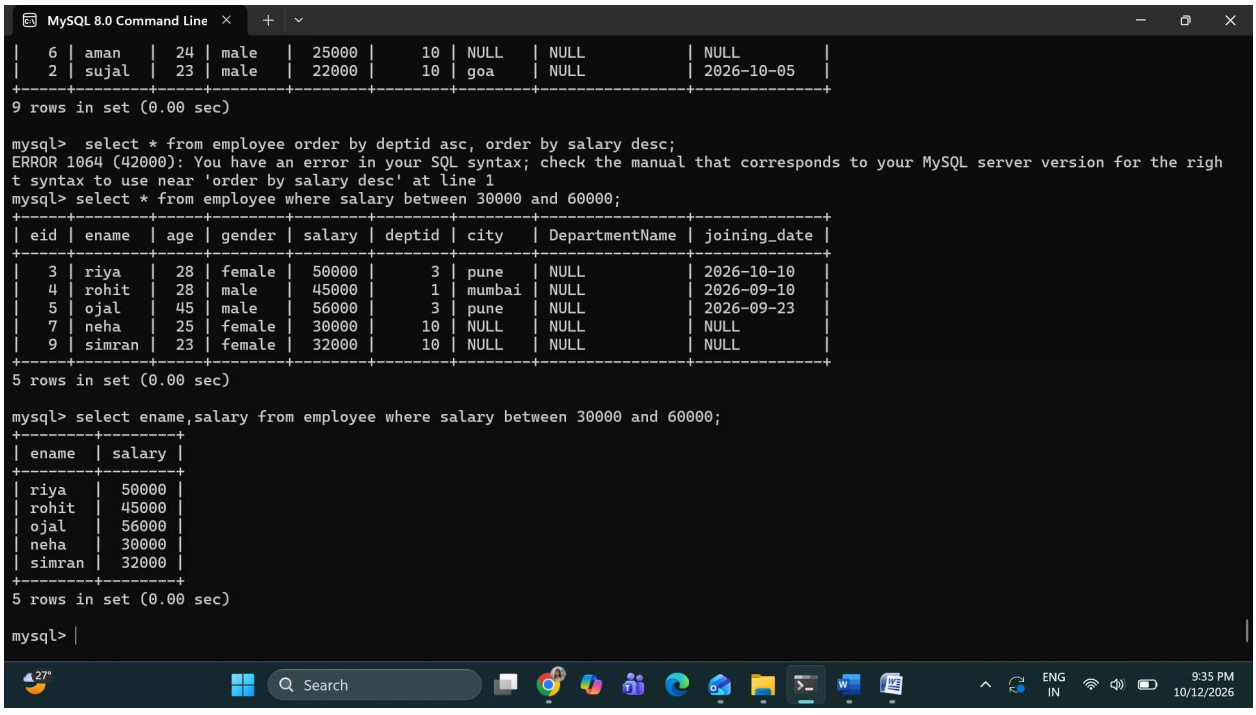
Q10.


MySQL 8.0 Command Line
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the righ
t syntax to use near 'by salary desc' at line 1
mysql> select salary from employee sort by salary desc;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the righ
t syntax to use near 'by salary desc' at line 1
mysql> select ename,salary from employee sort by desc;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the righ
t syntax to use near 'by desc' at line 1
mysql> select ename,salary from employee sort by salary desc;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the righ
t syntax to use near 'by salary desc' at line 1
mysql> select ename,salary from employee order by salary desc;
ename	salary
ojal	56000
riya	50000
rohit	45000
simran	32000
neha	30000
raj	28000
payal	27500
aman	25000
sujal	22000
9 rows in set (0.00 sec)
mysql>
Search
ENG
IN
9:10 PM
10/12/2026

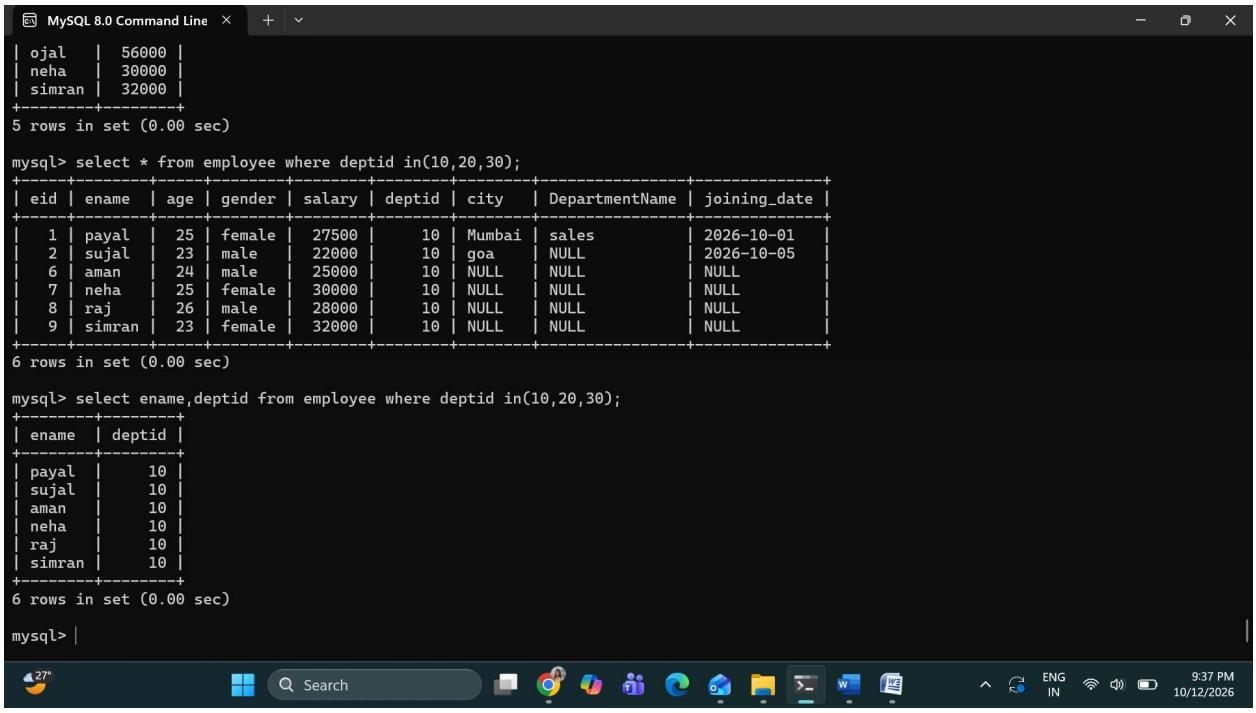
Q11.


MySQL 8.0 Command Line
mysql> select * from employee order by deptid , salary desc;
+-----+--------+-----+--------+--------+--------+--------+----------------+--------------+
| eid | ename | age | gender | salary | deptid | city | DepartmentName | joining_date |
+-----+--------+-----+--------+--------+--------+--------+----------------+--------------+
4	rohit	28	male	45000	1	mumbai	NULL	2026-09-10
5	ojal	45	male	56000	3	pune	NULL	2026-09-23
3	riya	28	female	50000	3	pune	NULL	2026-10-10
9	simran	23	female	32000	10	NULL	NULL	NULL
7	neha	25	female	30000	10	NULL	NULL	NULL
8	raj	26	male	28000	10	NULL	NULL	NULL
1	payal	25	female	27500	10	Mumbai	sales	2026-10-01
6	aman	24	male	25000	10	NULL	NULL	NULL
2	sujal	23	male	22000	10	goa	NULL	2026-10-05
+-----+--------+-----+--------+--------+--------+--------+----------------+--------------+
9 rows in set (0.00 sec)
mysql> select * from employee order by deptid asc, salary desc;
+-----+--------+-----+--------+--------+--------+--------+----------------+--------------+
| eid | ename | age | gender | salary | deptid | city | DepartmentName | joining_date |
+-----+--------+-----+--------+--------+--------+--------+----------------+--------------+
4	rohit	28	male	45000	1	mumbai	NULL	2026-09-10
5	ojal	45	male	56000	3	pune	NULL	2026-09-23
3	riya	28	female	50000	3	pune	NULL	2026-10-10
9	simran	23	female	32000	10	NULL	NULL	NULL
7	neha	25	female	30000	10	NULL	NULL	NULL
8	raj	26	male	28000	10	NULL	NULL	NULL
1	payal	25	female	27500	10	Mumbai	sales	2026-10-01
6	aman	24	male	25000	10	NULL	NULL	NULL
2	sujal	23	male	22000	10	goa	NULL	2026-10-05
+-----+--------+-----+--------+--------+--------+--------+----------------+--------------+
9 rows in set (0.00 sec)
Search
ENG
IN
9:30 PM
10/12/2026

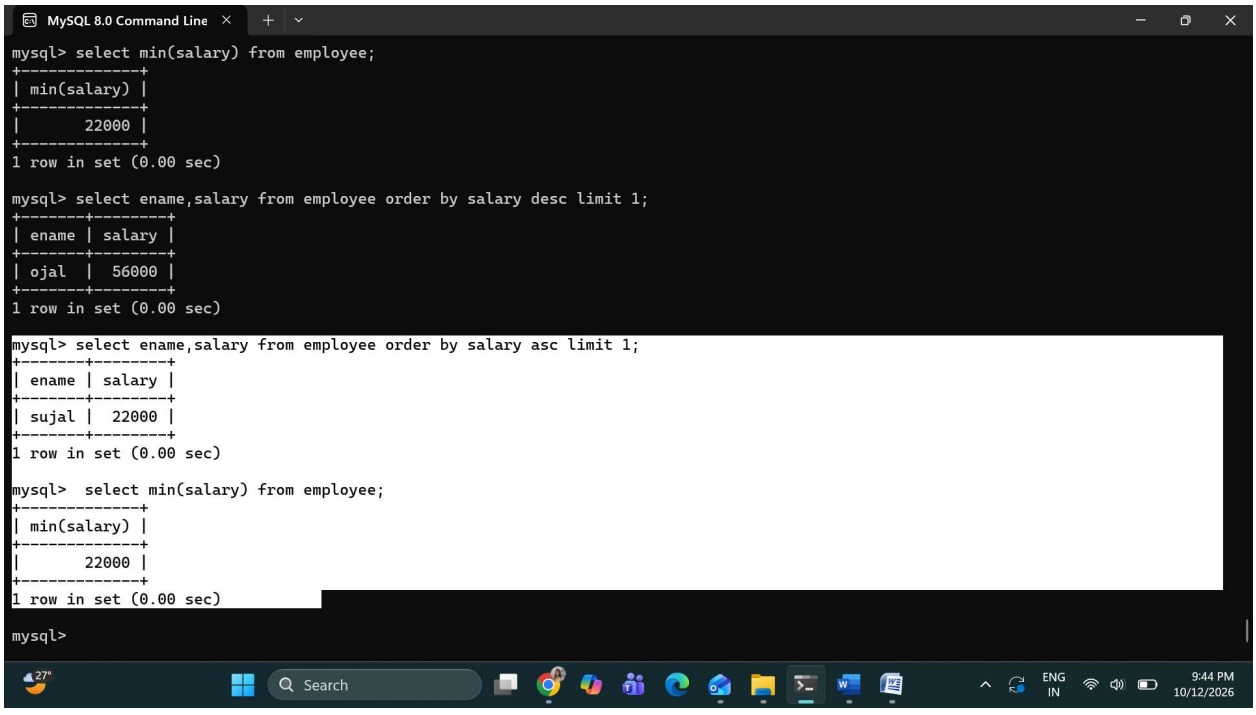
Q12.


MySQL 8.0 Command Line
| 6 | aman | 24 | male | 25000 | 10 | NULL | NULL | NULL |
| 2 | sujal | 23 | male | 22000 | 10 | goa | NULL | 2026-10-05 |
9 rows in set (0.00 sec)
mysql> select * from employee order by deptid asc, order by salary desc;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the righ
t syntax to use near 'order by salary desc' at line 1
mysql> select * from employee where salary between 30000 and 60000;
3	riya	28	female	50000	3	pune	NULL	2026-10-10
4	rohit	28	male	45000	1	mumbai	NULL	2026-09-10
5	ojal	45	male	56000	3	pune	NULL	2026-09-23
7	neha	25	female	30000	10	NULL	NULL	NULL
9	simran	23	female	32000	10	NULL	NULL	NULL
5 rows in set (0.00 sec)
mysql> select ename,salary from employee where salary between 30000 and 60000;
ename	salary
riya	50000
rohit	45000
ojal	56000
neha	30000
simran	32000
5 rows in set (0.00 sec)
mysql>
Search
ENG
IN
9:35 PM
10/12/2026

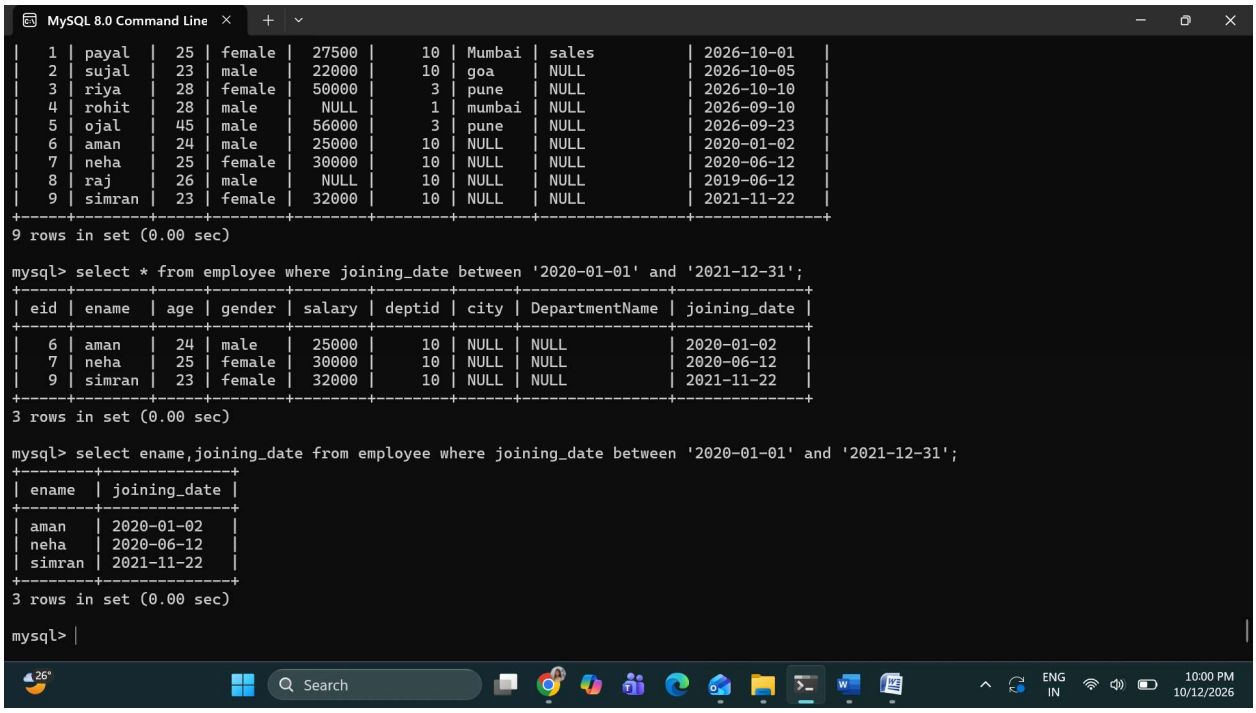
Q13.


MySQL 8.0 Command Line
ojal	56000
neha	30000
simran	32000
5 rows in set (0.00 sec)
mysql> select * from employee where deptid in(10,20,30);
eid	ename	age	gender	salary	deptid	city	DepartmentName	joining_date
1	payal	25	female	27500	10	Mumbai	sales	2026-10-01
2	sujal	23	male	22000	10	goa	NULL	2026-10-05
6	aman	24	male	25000	10	NULL	NULL	NULL
7	neha	25	female	30000	10	NULL	NULL	NULL
8	raj	26	male	28000	10	NULL	NULL	NULL
9	simran	23	female	32000	10	NULL	NULL	NULL
6 rows in set (0.00 sec)
mysql> select ename,deptid from employee where deptid in(10,20,30);
ename	deptid
payal	10
sujal	10
aman	10
neha	10
raj	10
simran	10
6 rows in set (0.00 sec)
mysql>
Search
ENG
IN
9:37 PM
10/12/2026

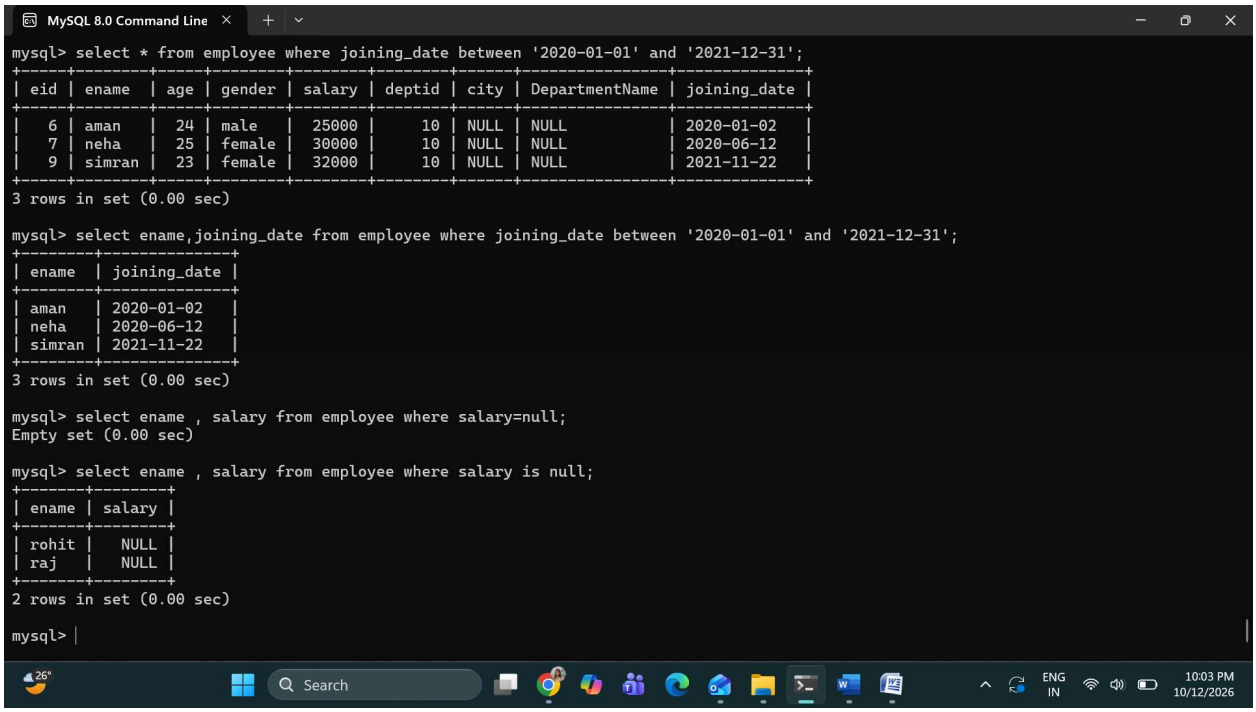
Q14.


MySQL 8.0 Command Line
mysql> select min(salary) from employee;
| min(salary) |
| 22000 |
1 row in set (0.00 sec)
mysql> select ename,salary from employee order by salary desc limit 1;
| ename | salary |
| ojal | 56000 |
1 row in set (0.00 sec)
mysql> select ename,salary from employee order by salary asc limit 1;
| ename | salary |
| sujal | 22000 |
1 row in set (0.00 sec)
mysql> select min(salary) from employee;
| min(salary) |
| 22000 |
1 row in set (0.00 sec)
mysql>
Search
ENG
IN
9:44 PM
10/12/2026

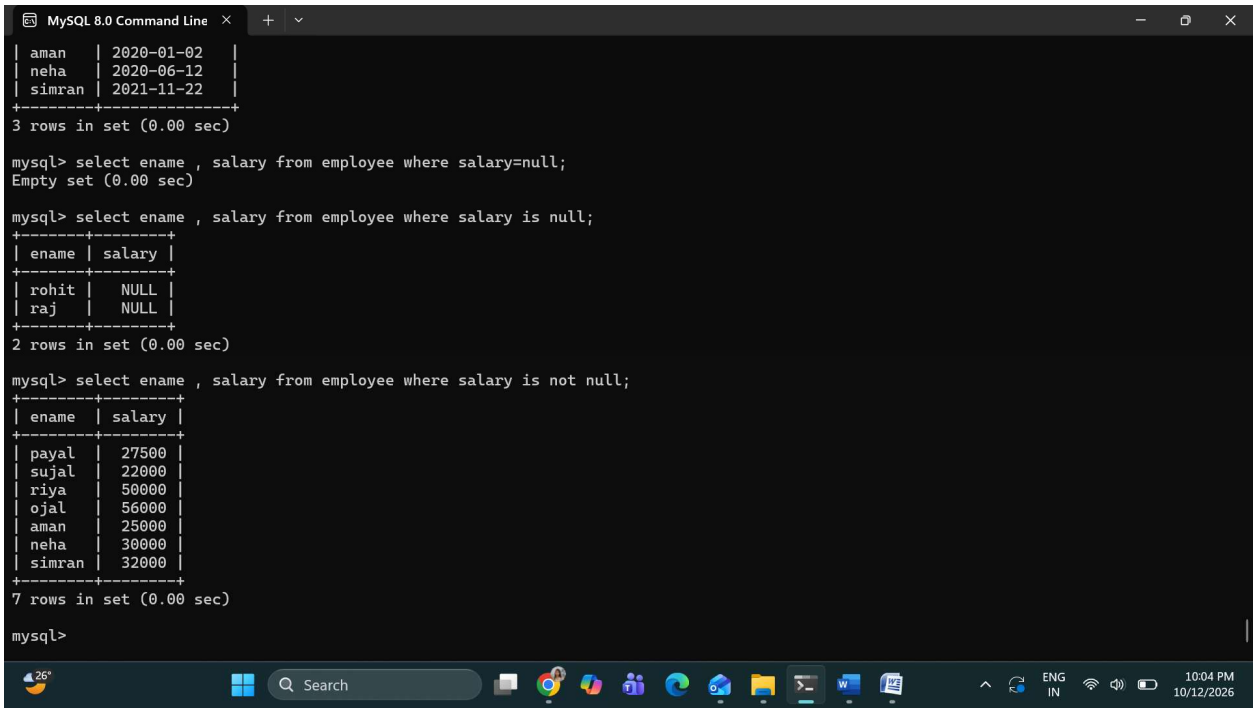
Q15.


MySQL 8.0 Command Line
1	payal	25	female	27500	10	Mumbai	sales	2026-10-01
2	sujal	23	male	22000	10	goa	NULL	2026-10-05
3	riya	28	female	50000	3	pune	NULL	2026-10-10
4	rohit	28	male	NULL	1	mumbai	NULL	2026-09-10
5	ojal	45	male	56000	3	pune	NULL	2026-09-23
6	aman	24	male	25000	10	NULL	NULL	2020-01-02
7	neha	25	female	30000	10	NULL	NULL	2020-06-12
8	raj	26	male	NULL	10	NULL	NULL	2019-06-12
9	simran	23	female	32000	10	NULL	NULL	2021-11-22
9 rows in set (0.00 sec)
mysql> select * from employee where joining_date between '2020-01-01' and '2021-12-31';
eid	ename	age	gender	salary	deptid	city	DepartmentName	joining_date
6	aman	24	male	25000	10	NULL	NULL	2020-01-02
7	neha	25	female	30000	10	NULL	NULL	2020-06-12
9	simran	23	female	32000	10	NULL	NULL	2021-11-22
3 rows in set (0.00 sec)
mysql> select ename,joining_date from employee where joining_date between '2020-01-01' and '2021-12-31';
ename	joining_date
aman	2020-01-02
neha	2020-06-12
simran	2021-11-22
3 rows in set (0.00 sec)
mysql>
Search
ENG
IN
10:00 PM
10/12/2026

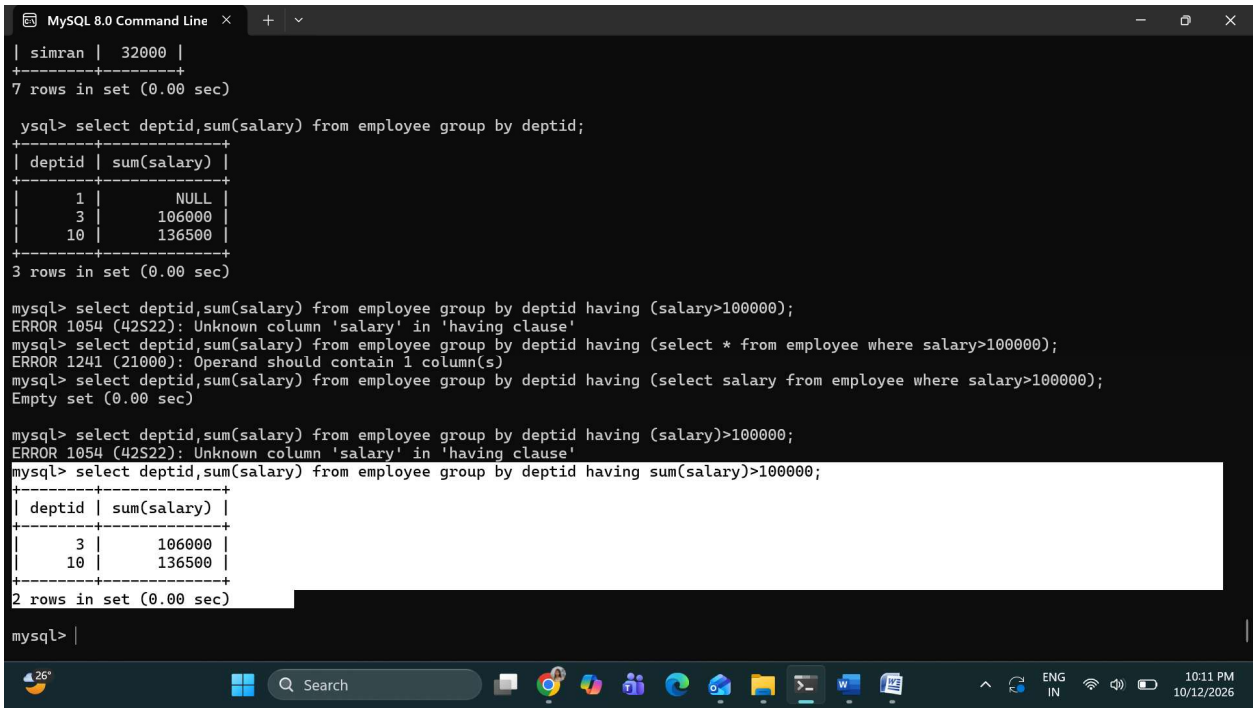
Q16.


MySQL 8.0 Command Line
mysql> select * from employee where joining_date between '2020-01-01' and '2021-12-31';
eid	ename	age	gender	salary	deptid	city	DepartmentName	joining_date
6	aman	24	male	25000	10	NULL	NULL	2020-01-02
7	neha	25	female	30000	10	NULL	NULL	2020-06-12
9	simran	23	female	32000	10	NULL	NULL	2021-11-22
3 rows in set (0.00 sec)
mysql> select ename,joining_date from employee where joining_date between '2020-01-01' and '2021-12-31';
ename	joining_date
aman	2020-01-02
neha	2020-06-12
simran	2021-11-22
3 rows in set (0.00 sec)
mysql> select ename , salary from employee where salary=null;
Empty set (0.00 sec)
mysql> select ename , salary from employee where salary is null;
ename	salary
rohit	NULL
raj	NULL
2 rows in set (0.00 sec)
mysql>
Search
ENG
IN
10:03 PM
10/12/2026

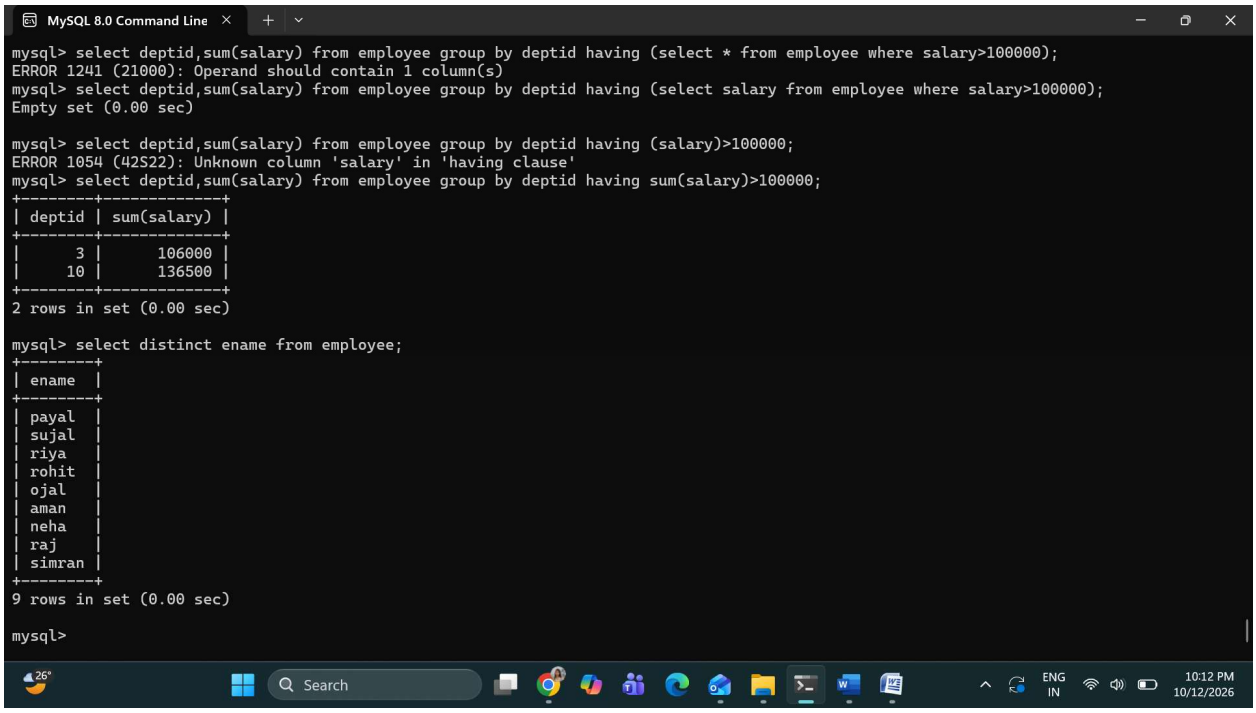
Q17.


MySQL 8.0 Command Line
aman	2020-01-02
neha	2020-06-12
simran	2021-11-22
+--------+--------------+
3 rows in set (0.00 sec)
mysql> select ename , salary from employee where salary=null;
Empty set (0.00 sec)
mysql> select ename , salary from employee where salary is null;
+-------+--------+
| ename | salary |
+-------+--------+
| rohit | NULL |
| raj | NULL |
+-------+--------+
2 rows in set (0.00 sec)
mysql> select ename , salary from employee where salary is not null;
+--------+--------+
| ename | salary |
+--------+--------+
payal	27500
sujal	22000
riya	50000
ojal	56000
aman	25000
neha	30000
simran	32000
+--------+--------+
7 rows in set (0.00 sec)
mysql>
Search
ENG
IN
10:04 PM
10/12/2026

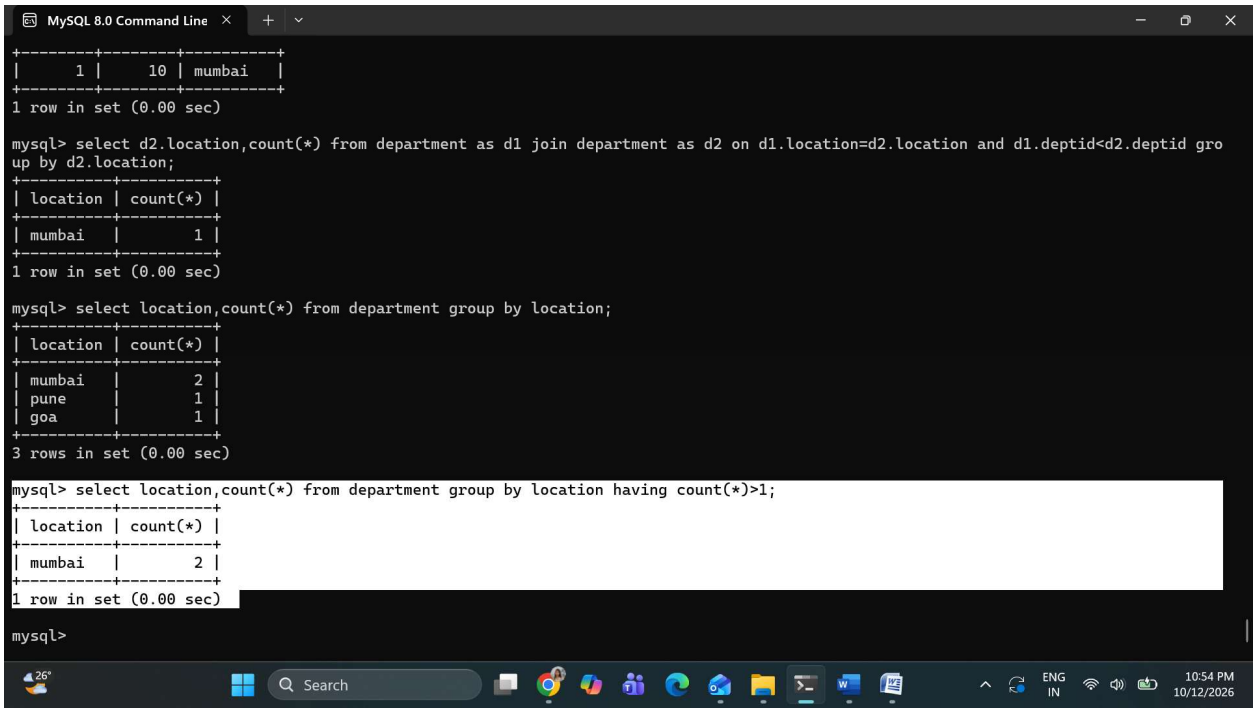
Q18.


MySQL 8.0 Command Line
| simran | 32000 |
+--------+--------+
7 rows in set (0.00 sec)
ysql> select deptid,sum(salary) from employee group by deptid;
+--------+-------------+
| deptid | sum(salary) |
+--------+-------------+
1	NULL
3	106000
10	136500
+--------+-------------+
3 rows in set (0.00 sec)
mysql> select deptid,sum(salary) from employee group by deptid having (salary>100000);
ERROR 1054 (42S22): Unknown column 'salary' in 'having clause'
mysql> select deptid,sum(salary) from employee group by deptid having (select * from employee where salary>100000);
ERROR 1241 (21000): Operand should contain 1 column(s)
mysql> select deptid,sum(salary) from employee group by deptid having (select salary from employee where salary>100000);
Empty set (0.00 sec)
mysql> select deptid,sum(salary) from employee group by deptid having (salary)>100000;
ERROR 1054 (42S22): Unknown column 'salary' in 'having clause'
mysql> select deptid,sum(salary) from employee group by deptid having sum(salary)>100000;
+--------+-------------+
| deptid | sum(salary) |
+--------+-------------+
| 3 | 106000 |
| 10 | 136500 |
+--------+-------------+
2 rows in set (0.00 sec)
mysql>
Search
ENG
IN
10:11 PM
10/12/2026

Q19.


MySQL 8.0 Command Line
mysql> select deptid,sum(salary) from employee group by deptid having (select * from employee where salary>100000);
ERROR 1241 (21000): Operand should contain 1 column(s)
mysql> select deptid,sum(salary) from employee group by deptid having (select salary from employee where salary>100000);
Empty set (0.00 sec)
mysql> select deptid,sum(salary) from employee group by deptid having (salary)>100000;
ERROR 1054 (42S22): Unknown column 'salary' in 'having clause'
mysql> select deptid,sum(salary) from employee group by deptid having sum(salary)>100000;
deptid	sum(salary)
3	106000
10	136500
2 rows in set (0.00 sec)
mysql> select distinct ename from employee;
ename
payal
sujal
riya
rohit
ojal
aman
neha
raj
simran
9 rows in set (0.00 sec)
mysql>
Search
ENG
IN
10:12 PM
10/12/2026

Q20.


MySQL 8.0 Command Line
| 1 | 10 | mumbai |
1 row in set (0.00 sec)
mysql> select d2.location,count(*) from department as d1 join department as d2 on d1.location=d2.location and d1.deptid<d2.deptid gro
up by d2.location;
| location | count(*) |
| mumbai | 1 |
1 row in set (0.00 sec)
mysql> select location,count(*) from department group by location;
location	count(*)
mumbai	2
pune	1
goa	1
3 rows in set (0.00 sec)
mysql> select location,count(*) from department group by location having count(*)>1;
location	count(*)
mumbai	2
1 row in set (0.00 sec)
mysql>
Search
ENG
IN
10:54 PM
10/12/2026